

On Research Designs

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1. Introduction

Research design provides a framework for the collection and analysis of data, shaping how research questions are addressed and how conclusions are drawn. The process begins with defining a clear research question, which serves as the foundation for a study. Once the research question is established, researchers develop a research design; the overall plan that determines how the question will be addressed. Only after the design is finalized do researchers begin to collect and analyze data, guided by the decisions made in the design phase.

Research design involves decisions about the study's structure, guiding the use of research methods and ensuring the study aligns with its objectives. Importantly, research design is distinct from research methods: while methods are specific techniques for collecting data (e.g., surveys or interviews), research design reflects broader choices about the overall approach, priorities, and organization of the research.

A helpful analogy is the construction of a building (cf. de Vaus 2001). The research design is similar to deciding what type of building we want to construct—whether it will be a skyscraper, a house, or a factory. This decision informs every other aspect of the process, from the methods relied upon, type of data collected etc. The primary function of a research design is to ensure that the research question can be answered, given the methods, data, and overall structure of the study. A well-thought-out design aligns these elements to produce credible and meaningful results.

Bell et al. (2022) argues that different research designs, assign different levels of importance to the following **four dimensions**:

- “expressing causal connections between variables;
- generalizing to larger groups of individuals than those actually forming part of the investigation;
- understanding behaviour and the meaning of that behaviour in its specific social context;
- having a temporal (i.e., over time) appreciation of social phenomena and their interconnections.” (Bell et al. 2022, p. 40).

In this chapter, we explore three major research designs—**cross-sectional, longitudinal, and case studies**—each reflecting distinct priorities regarding these dimensions (see Bergenholtz et al. 2024a for information on **experiments**, the fourth research design). Understanding their strengths, weaknesses, and usefulness will enable critical assessment of research studies and informed selection of designs in future research.

2. Cross-Sectional Design

A cross-sectional design captures data at a single point in time, providing a snapshot of relationships between variables. This design is particularly effective for exploring **generalizations to larger groups of individuals** (cf. the 4 bullet points in section 1 just above) and for identifying correlations between variables. However, it is less useful for understanding behavior in specific

contexts or capturing temporal changes over time. For instance, instead of asking why a specific individual chose a particular education or job, a cross-sectional design focuses on identifying the **average** relation between different types of education (while potentially controlling for other factors) and salary.

Illustrating a cross-sectional design

To visualize cross-sectional research, imagine an Excel sheet where each row represents an individual (e.g., employees or students), and each column represents a variable, such as education level, salary, years of experience, job satisfaction, and industry type. Each cell contains one specific data point for each variable (e.g. education and salary) for an individual, creating an extensive dataset for analysis.

Respondent	Education	Salary / Month	Years of experience	Job Satisfaction (1-5)	Industry type
1	BA	42.000	3	3	IT
2	MA	52.000	6	4	Finance
3	College	32.000	5	2	Manufacturing
4	BA	48.000	2	5	Finance
5	BA	38.000	8	5	Manufacturing
6	MA	60.000	10	3	IT
7	PhD	65.000	5	4	Academia
...	...	...	...	...	...

Table 1: Illustrating cross-sectional data.

Note that while cross-sectional designs are defined by relying on data on one point in time, one can imagine that the data collection process lasts longer. For example, imagine a researcher who collects data on how much people exercise over a period of a month. The data is collected by asking people once a week how much they exercised, but in the end the researcher only keeps and uses one data point for each participant (how much they exercised, on average across the weeks). This approach would be cross-sectional, since only one data point is available, and the focus is on summarizing relationships across variables and participants, not on investigating changes over time within that month. Also note that one does not have to collect the data oneself; one can use data others have collected.

Strenghts and weaknesses

A well designed cross-sectional study in a fairly large sample, where the researcher has controlled for known, important confounding factors, can provide a **suggestion for a strong causal link** - note, not a strong suggestion for a causal link. However, there is always a risk that any observed relationships between variables might be influenced by unmeasured factors (omitted variable bias, see Bergenholtz 2025e). For example, if a study finds a correlation between exercise and mood, it might try to control for a range of factors, such as weight, diet, stress, socioeconomic factors (etc.), but it is impossible to know for certain if those factors have been measured precisely or if other factors (genetics for example) actually constitute the causal factor. Or, following Isager (2023), it might be that it is actually a better mood that leads to more exercise. See Bergenholtz 2025e for more on these kinds of challenges to cross-sectional studies.

Generally, cross-sectional designs prioritize **external validity** (the ability to generalize findings to larger populations) over **internal validity** (confidence in finding causal relationships in this particular dataset). This approach also reflects specific assumptions about how knowledge is created in social science. By focusing on patterns across many cases rather than deeply analyzing specific instances (as a case study, see section 4), cross-sectional designs emphasize identifying broad, generalizable trends. This aligns with the goal of discovering consistent relationships across populations rather than understanding the unique details of individual cases.

3. Longitudinal Design

A longitudinal design involves collecting data from the same entities (e.g. individuals, groups or organizations) at multiple points in time. This approach allows researchers to observe changes and trends, offering insights into how variables develop or relate to each other over time. Unlike cross-sectional studies, which capture a single moment in time, longitudinal studies follow individuals, groups, or organizations over an extended period, tracking their progression or change. There are two main types of longitudinal studies:

- **Panel Studies:** These involve repeatedly collecting data from the same set of participants over time. The defining feature is tracking individual-level change, though some participant dropout typically occurs. **Example:** Following the same 100 employees to monitor their exercise habits and mood over a year, even if some participants eventually leave the study.
- **Cohort Studies:** These involve tracking a group that shares a common characteristic (e.g., starting the same job, graduating in the same year) and observing their development over time. The focus is on group-level trends, with different individuals from the cohort potentially being surveyed at each time point. **Example:** Studying employees hired in 2020 by surveying different members of this hiring cohort each quarter to understand career progression patterns, without necessarily reaching the same individuals each time.

Longitudinal research primarily addresses two dimensions from the introduction: **expressing causal connections between variables** and **having a temporal (over time) appreciation of phenomena**. By observing the sequence of events, longitudinal studies provide stronger evidence for causality than cross-sectional designs. For example, if increased exercise consistently emerges before improved mood in several time points, this suggests a directional relationship.

Respondent	Time Point	Exercise Frequency (Days/Week)	Mood Rating (1-5)
1	T1	2	3
1	T2	4	4
1	T3	5	5
2	T1	0	2
2	T2	1	3
2	T3	2	4
3	T1	3	4
3	T2	3	4
3	T3	4	5
4	T1	1	2
4	T2	2	3
4	T3	3	4
...	...	...	...

Table 2: Illustrating a longitudinal design. Note there are multiple data points for each individual.

While this suggests a relationship, the evidence is not conclusive. Maybe measurements of mood are somewhat delayed. Or, maybe an omitted factor—such as participation in a wellness program that encourages both exercise and stress management—could be influencing both exercise and mood. Without accounting for such omitted variables or measurement challenges, the study might not correctly assess the direct link between exercise and mood.

Strengths and Weaknesses

Longitudinal designs often prioritize **internal validity** by tracking changes over time and establishing temporal sequences. By following such changes, there is an opportunity to measure if a change in the independent variable happens before a change in the dependent variable. This is something a cross-sectional study can not ensure. However, longitudinal studies typically focus on specific populations and might involve smaller samples, which can limit **external validity**, making generalization to broader populations more challenging. Furthermore, the risk of omitted variable bias is still in play, as in cross-sectional studies. Just because one has data over time, does not mean that one has eliminated the risk of overlooking an important variable. A few additional methodological challenges deserve attention in longitudinal research:

- **Time-Related Effects:** When conducting research over time, several factors can affect results independently of what is being studied. Natural developments, changing social contexts and people dropping out of a study can create biases that are difficult to identify. **For example:** Imagine one is studying a situation where employees are getting specific on-the-job training, and one is interested in seeing the effects of this over time. If employees demonstrate an improved job performance over time, this might be due to general on-the-job experience one

would have gotten anyways, rather than the specific training programs being studied (the natural development). Or, shifting cultural norms could affect how survey questions are answered over time, meaning what looks like a change in performance, could be due to how performance ratings are created and perceived by employees and managers (changing social contexts). Finally, employees who quit might be systematically different from those who remain. Hence, if over time one sees a change in a relevant variable, this might simply be due to some people (low performers for example) dropping out (called attrition bias). Researchers must account for these time related factors and participation patterns to be sure to make valid conclusions.

Summing up

Longitudinal research recognizes that people, behaviors, and societies are dynamic—they change over time. This approach emphasizes the importance of assessing these changes, not only to understand behaviors better but also to improve our ability to address causality. By examining patterns and developments across time, longitudinal studies provide insights into both the mechanisms of change and the factors that drive them.

4. Case study Design

A case study design focuses on an in-depth exploration of a specific, bounded system, such as an organization, event, individual, or process. Rather than attempting to understand averages or general trends across populations, case studies aim to uncover the nuanced dynamics and unique characteristics of the subject under investigation. Referring back to the four bullet points in the introduction, this design is **aimed at understanding behavior and the meaning of that behavior in its specific social context**. By diving deeply into a particular case, researchers create insights into the particular factors and interactions that shape what happens in the case context.

Illustrating a Case Study Design

Imagine a case study examining why implementing chatbots in a customer service organization might be challenging. The researcher could use a several methods—such as observing chatbot usage during customer interactions, conducting interviews with employees and managers, distributing surveys to gauge employee attitudes, and analyzing company performance data. The aim would be to investigate how various factors (e.g. organizational culture, employee skills, and existing workflows) shape the adoption of the chatbot system.

To illustrate the nature of a case study, one can contrast it with the cross-sectional design: **The kinds of research questions asked in cross-sectional and case study designs are very different** in focus and aim. In a cross-sectional design, questions typically aim to identify general patterns or correlations across a population at a single point in time. For example: “What is the relationship between job satisfaction and productivity across employees in a customer service industry?” The emphasis is on general and statistical patterns.

In contrast, a case study design focuses on deep, context-specific questions that seek to understand the unique interactions and processes within a single or small set of cases. For instance: “How do organizational culture and employee attitudes shape the adoption of chatbots in a specific customer service company?” These questions emphasize depth, context, and the meaning of behaviors or events, rather than broad patterns across many data points.

Multiple Case Studies: In some instances, researchers might adopt a multiple case study design to compare and contrast findings across different contexts (Eisenhardt 1989). Cases are typically selected not for statistical representativeness but for their ability to highlight theoretical differences or similarities (Flyvbjerg 2006). For example, when studying chatbot implementation, organizations might be chosen to represent different levels of technological readiness or different organizational cultures. Single case studies provide an opportunity to focus deeply on a particular context, offering rich and nuanced insights into the unique dynamics of that case. In contrast, multiple case studies (e.g., examining 4 or 8 cases) focus less on the individual uniqueness of each context and more on identifying variations or similarities across cases (Dyer and Wilkins 1991). This approach is arguably somewhat related to the aim of cross-sectional designs, as it emphasizes comparative analysis and generalization across contexts while still using the depth of qualitative insights from each case. By comparing these cases, researchers can identify patterns, refine theories, and draw broader conclusions, thus trying to capture both the cases' depth and broader applicability (Eisenhardt 1989).

Longitudinal: Case studies can also be longitudinal, tracking the process of adoption and integration over time. For example, the researcher might study the organization during the planning, initial implementation, and post-implementation phases to understand how challenges change and how the organization responds.

Strengths and Weaknesses

Case studies do well in providing rich, detailed insights into specific phenomena, especially when the research question revolves around “how” or “why” something happens. They are particularly valuable for understanding complex, context-specific interactions that might be missed in broader quantitative, e.g. cross-sectional, studies. Case study research is usually related to a kind of explanation that focuses on understanding as a primary concern, rather than causality (see Bergenholtz 2025f).

However, their findings often have limited **external validity**, as the depth and specificity of the analysis may not transfer to other contexts. Case studies can also be resource-intensive and subject to researcher bias, as the interpretation of data often depends on the researcher's perspective and approach.

Summing up

This design reflects an assumption that knowledge is created through deep, context-specific exploration. By focusing on the unique details of a case, researchers aim to understand the phenomenon itself and the interactions that shape it within its specific case context. Case studies are not designed to generalize findings or uncover universal laws, but they can generate valuable insights and inform broader theories, policies, or practices.

5. Comparing Research Designs

How to identify different research designs

To determine the appropriate research design, or to identify which design is being used in a study, consider the focus and structure of the research. If the study collects data from individuals at a single point in time to investigate patterns or correlations—such as surveying employees' job satisfaction and performance in one week—it is a **cross-sectional design**. If the study instead collects data from the same individuals at multiple points over time—such as measuring employees'

job satisfaction before, during, and after implementing a new policy—it is a **longitudinal design**, which emphasizes observing change or development. When the study focuses on a specific case (or few cases), such as understanding why a particular company struggled to adopt a new technology, and integrates methods like interviews, observations, surveys, or other perspectives, it is a **case study design**, prioritizing contextual understanding. If the study involves manipulating a variable—such as giving randomly selected employees access to a training program while not giving it to other employees to measure differences in performance—it is an **experimental design**, which focuses on isolating causal relationships through control and randomization (see Bergenholtz 2024a).

Combining research designs

Research designs aren't always used in isolation. Studies often combine multiple designs to take advantage of their complementary strengths. For instance, a researcher might use cross-sectional surveys across multiple organizations to identify average patterns while conducting in-depth case studies of selected companies to understand the mechanisms that drive those patterns. One could also do a longitudinal field-experiment that even includes a case study where one observes how the treatment intervention shapes human behavior.

On generalization (external validity)

Generalizability differs significantly across research designs and is closely tied to their assumptions about causality and the nature of their findings.

Cross-sectional and experimental designs often aim for **statistical generalizability**, arguing that findings from a sample (if representative) can be applied to a larger population. These designs operate on the assumption that causality is stable and predictable. For example, an experiment might test whether offering bonuses increases productivity and, if successful, claim that this relationship holds universally across similar contexts. Repeated replication of such studies—redoing them in either the same or different contexts—further enhances the reliability of their conclusions.

In contrast, case studies aim for **theoretical generalizability**, often referred to as transferability. This approach asks whether insights from a particular context can inform understanding in similar settings. For instance, a case study investigating the introduction of a bonus scheme in a single company might uncover how factors like organizational culture or employee expectations are important for the scheme's implementation and outcome. While these findings are not statistically generalizable, they can offer valuable lessons for other organizations with similar characteristics.

Basic philosophy of science assumptions

These differences in generalizability reflect deeper **epistemological and ontological** distinctions (see Guba 1990, Bergenholtz 2024f, curriculum in later lessons). Cross-sectional and experimental designs are typically associated with **positivist approaches**, prioritizing objectivity and the search for existing, underlying, general laws of businesses practices and outcomes. Case studies, on the other hand, generally reflect **constructivist perspectives**, emphasizing the particularity of individual contexts and the idea that researchers actively construct an understanding of a context. These philosophical underpinnings influence not only how research is conducted but also how its findings are interpreted and applied.

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